SoftAND* and SoftThresh

Residual misspecification families for the hybrid / DR study
A self-contained explanation (math + intuition + locked specs)

OPC project

November 11, 2026

Abstract

This note explains the two *latent residual* families used alongside $R^\star$ in the confirmatory misspecification grid: **SoftAND*** and **SoftThresh** (locked τ=1.5). It is written for a reader who has not followed the implementation chat: what problem they solve, how they are built from embeddings, how they differ from each other and from $R^\star$, which hyperparameters are locked in the runs, and how the dial γ shows up in realized logit variance / reward ratios.

Contents

1 Where these families sit in the experiment **2**
1.1 Truth logit with a residual dial . . . 2
1.2 Why SoftAND* / SoftThresh exist . . . 2

2 Shared scaffolding **2**
2.1 Facet projections . . . 2
2.2 Bilinear perpendicular ("⊥") . . . 2
2.3 Unit-SD (norm=v2) . . . 3

3 SoftAND* — thresholded per-facet concordance **3**
3.1 Intuition . . . 3
3.2 Formula (locked) . . . 3
3.3 Locked residual-spec string . . . 3
3.4 What it is / is not . . . 3

4 SoftThresh ($\tau = 1.5$) — soft threshold on z_2 **3**
4.1 Intuition . . . 3
4.2 Formula (locked) . . . 4
4.3 Locked residual-spec string . . . 4
4.4 Role of τ . . . 4

5 Side-by-side comparison **4**

6 Dial calibration (MovieLens, bias high) **4**

7 How this shows up in learner results **5**

8 Code & tags **5**

9 One-line takeaways **5**

1 Where these families sit in the experiment

1.1 Truth logit with a residual dial

Users x and items a have fixed BPR (or mlsize) embeddings. The environment click probability is $q = \sigma(\ell_\gamma)$ with

$$\ell_\gamma(x,a) = \underbrace{b + u_x^\top v_a}_{\text{“GT” bilinear / affine-bilinear}} + \gamma \cdot \alpha \cdot \phi(x,a), \tag{1}$$

where ϕ is standardized to unit SD over (x,a) pairs ($\mathrm{sd}(\phi) = 1$), α is a fixed scale from the protocol, and $\gamma \geq 0$ is the **misspecification dial**.

- $\gamma = 0$: well-specified bilinear world (residual off).
- Mid γ (e.g. 0.5–2): residual competes with the GT channel.
- Extreme γ (e.g. 10–100): residual dominates ranking.

A global offset $b'(\gamma)$ is recalibrated so a logger reference CTR stays near 5% as γ changes (same protocol as $R^\star$).

1.2 Why SoftAND* / SoftThresh exist

$R^\star$ (even tanh warp of a second facet product) is one inductive bias: a *separable* second interaction channel that is approximately orthogonal to linear scores because of evenness. SoftAND* and SoftThresh ask whether the hybrid / DR story still holds under **different residual geometries**:

- SoftAND*: *per-facet* “both sides must fire” concordance, then strip anything bilinear can already represent.
- SoftThresh: a *scalar* soft threshold on a second-index score — sparse “high second-index” bonuses that are *not* finite-rank separable.

Learners (NLL, OPC/DR, hybrid_seq) never see ϕ as a feature; they only see logged clicks from $q(\ell_\gamma)$. Misspecification means their assumed scoring class (linear in the same embeddings) is wrong for large γ.

2 Shared scaffolding

2.1 Facet projections

Draw (once per seed / `proj_tag`) random projections so each user and item has m facet coordinates $t_{u,k}, t_{a,k}$ (z-scored). Locked grid uses

$$m = 4, \texttt{proj_tag=default}, \texttt{norm=v2}.$$

The same (t_u, t_a) draw is shared across SoftAND* / SoftThresh / Cosine / PL on a given tag so family differences are not confounded with projection noise.

2.2 Bilinear perpendicular (“⊥”)

After building a raw residual score ϕ_{raw}, SoftAND* and SoftThresh apply an explicit projection that removes the best affine-bilinear fit in the embedding features (and intercepts). Intuitively:

$$\phi \;=\; \text{unit-sd}(\phi_{\text{raw}} - \Pi_{\text{bilinear}}(\phi_{\text{raw}}))\,. \tag{2}$$

So the residual that multiplies $\gamma\alpha$ is “what a clean bilinear model cannot already soak up.” $R^\star$ mostly relied on evenness for a similar effect; SoftAND* / SoftThresh enforce $\perp$ explicitly.

2.3 Unit-SD (norm=v2)

After $\perp$, rescale so the catalog pair-SD of ϕ is 1. Then the designed residual contribution is exactly $\gamma\alpha\phi$ in SD units (SoftAND* / SoftThresh track sd_ratio $\approx \gamma$ empirically; $R^\star$ overshoots $\approx 2.4\times$).

3 SoftAND* — thresholded per-facet concordance

3.1 Intuition

Think of each facet k as a soft "topic light." SoftAND* gives reward mass when *the same* facet is above a threshold on *both* the user and the item — an AND-like concordance — then averages over facets.

3.2 Formula (locked)

With softplus $\mathrm{sp}(z) = \log(1 + e^z)$ and locked $c = 1$, $s = \texttt{gate_s} = 0.25$:

$$g_{u,k} = \mathrm{sp}\left(\frac{t_{u,k} - c}{s}\right), \qquad h_{a,k} = \mathrm{sp}\left(\frac{t_{a,k} - c}{s}\right), \tag{3}$$

$$\phi_{\mathrm{raw}}(u, a) = m^{-1/2} \sum_{k=1}^{m} g_{u,k}\, h_{a,k}. \tag{4}$$

Then double-center the factor sides, apply $\perp$, then unit-SD (v2).

Implementation sketch (`utils/residual_env.py`): factorized form $f_u \leftarrow m^{-1/2}\mathrm{sp}((t_u - c)/s)$, $f_a \leftarrow \mathrm{sp}((t_a - c)/s)$, so $\phi_{\mathrm{raw}} = f_u^\top f_a$ before centering / $\perp$.

3.3 Locked residual-spec string

```
m=4,c=1,gate_s=0.25,proj_tag=default,norm=v2
```

(KuaiRec mlsize cells may append `catalog_amp=0.75`.)

3.4 What it is / is not

- **Is:** multiplicative gate on co-activated facets; still *separable* before $\perp$ (rank-$\leq m$ product).
- **Is not:** an even warp of a second bilinear index ($R^\star$); not a threshold on the *sum* of facet products (that is SoftThresh).
- **vs $R^\star$:** SoftAND* uses one-sided softplus gates (odd-ish / threshold behavior), not $\tanh(s(t^2 - 1))$.

4 SoftThresh ($\tau = 1.5$) — soft threshold on z_2

4.1 Intuition

First form a second bilinear index from the same facets,

$$z_2(u, a) = m^{-1/2} \sum_{k=1}^{m} t_{u,k}\, t_{a,k}. \tag{5}$$

Standardize to $\tilde{z}_2 = (z_2 - \mu)/\sigma$ over the catalog. SoftThresh adds a *sparse bonus* only when $\tilde{z}_2$ exceeds a threshold τ — softplus so the map is smooth for gradients / SD math, but still "mostly off" below τ.

4.2 Formula (locked)

With $\tau = 1.5$, $s_{\text{th}} = 0.25$:

$$\phi_{\text{raw}}(u, a) = \text{sp}\left(\frac{\tilde{z}_2(u, a) - \tau}{s_{\text{th}}}\right). \tag{6}$$

Then $\perp$ against affine-bilinear features and unit-SD. SoftThresh is stored as a catalog residual (not a low-rank $f_u^\top f_a$ after the softplus), because softplus-of-a-sum is **not** finite-rank separable.

4.3 Locked residual-spec string

`m=4,tau=1.5,s_th=0.25,proj_tag=default,norm=v2`

(Optional `catalog_amp=0.75` on KuaiRec mlsize.)

4.4 Role of τ

Larger $\tau \Rightarrow$ rarer "on" pairs $\Rightarrow$ more spike-like residual before unit-SD (unit-SD then stretches the sparse mass back to SD 1). Locked confirmatory value: $\tau = 1.5$.

5 Side-by-side comparison

	SoftAND*	SoftThresh τ=1.5
Where nonlinearity sits	Per facet, then product	On scalar $\tilde{z}_2$
AND semantics	Same-facet co-activation	"Second index large" (any facets)
Separable after ψ?	Yes before $\perp$	No (softplus of a sum)
$\perp$	Yes	Yes
Tracks sd_ratio $\approx \gamma$?	Yes	Yes
Typical mid-γ var share of ϕ	$\approx 20\%$ at γ=0.5	$\approx 20\%$ at γ=0.5

Contrast with $R^\star$ at the same nominal γ=0.5: realized sd_ratio ≈ 1.19 and var_share$_\phi \approx 59\%$ — $R^\star$ is a *louder* dial at equal γ. That is why SoftAND* / SoftThresh often need slightly higher γ before hybrid clearly beats NLL, while still matching the designed SD dial.

6 Dial calibration (MovieLens, bias high)

Uniform (x, a) measurement (seeds 300–319). Designed sd$(\phi) = 1$. Selected rows:

family	γ	sd_ratio	var_share$_\phi$	reward_ratio	CTR lift$_\phi$
SoftAND*	0.5	0.50	19.9%	1.14	+0.33pt
SoftAND*	1	1.00	49.8%	1.44	+0.89pt
SoftAND*	2	1.99	79.9%	2.43	+1.86pt
SoftAND*	100	99.6	100%	$\gg 1$	+4.25pt
SoftThresh	0.5	0.50	20.0%	1.17	+0.39pt
SoftThresh	1	1.00	50.0%	1.57	+1.05pt
SoftThresh	2	2.00	80.0%	3.16	+2.17pt
SoftThresh	100	100.1	100%	$\gg 1$	+4.25pt
$R^\star$ (ref)	0.5	1.19	58.7%	1.72	+1.17pt

Reading. SoftAND* / SoftThresh are faithful to the nominal dial. At extreme γ, $E[q_{\text{GT}}] \to 0$ so raw reward ratios explode — prefer var_share$_\phi$ / share-of-$E[q]$ for interpretation. Full table: `hybrid_by_cell/REWARD_RATIOS.md`.

7 How this shows up in learner results

Primary estimands (exact-greedy CTR %), paired over seeds:

$$\Delta_{\mathrm{H,NLL}}(\gamma) = V(\pi_\gamma^{\mathrm{hybrid}\, f=0.5}) - V(\pi_\gamma^{\mathrm{NLL}}), \tag{7}$$

$$\mathrm{DiD} = \Delta(\gamma) - \Delta(0). \tag{8}$$

On MovieLens (n=40), SoftThresh typically shows a clear hybrid win from about $\gamma \geq 0.5$ (CI LB$>$ 0), SoftAND* from about $\gamma \geq 0.75$, with both families large and positive at $\gamma \in \{2, 4, 10, 100\}$. Well-spec γ=0 cells stay near zero / slightly negative for hybrid vs NLL (small cost of the DR phase when the world is bilinear). See `hybrid_results_by_setup.pdf` for per-setup tables (dataset×family×γ) with CIs and bolded best methods.

8 Code & tags

- Implementation: `utils/residual_env.py` (`kind=softand` / `softthresh`).
- Family overview (incl. $R^\star$, foils): `FAMILIES_EXPLAINED.md`.
- Full-grid run tags: `nlres_grid_{ds}_softand_g{`γ`}`, `nlres_grid_{ds}_softthresh_g{`γ`}`.
- Earlier ML locks: `nlres_h1_softand_g*`, `nlres_h1_st15_g*`, `nlres_h1b_softand_g*` (cfg-clean seed extend).

9 One-line takeaways

1. SoftAND* = soft thresholded **same-facet AND**, then strip bilinear, unit-SD.
2. SoftThresh = soft threshold on a **second bilinear index** z_2, then strip bilinear, unit-SD.
3. Both track sd_ratio $\approx \gamma$; $R^\star$ is louder at the same nominal γ.
4. They are stress tests that hybrid NLL→DR is not an $R^\star$-only artifact.